SQL JOIN – Multi-Level Lab

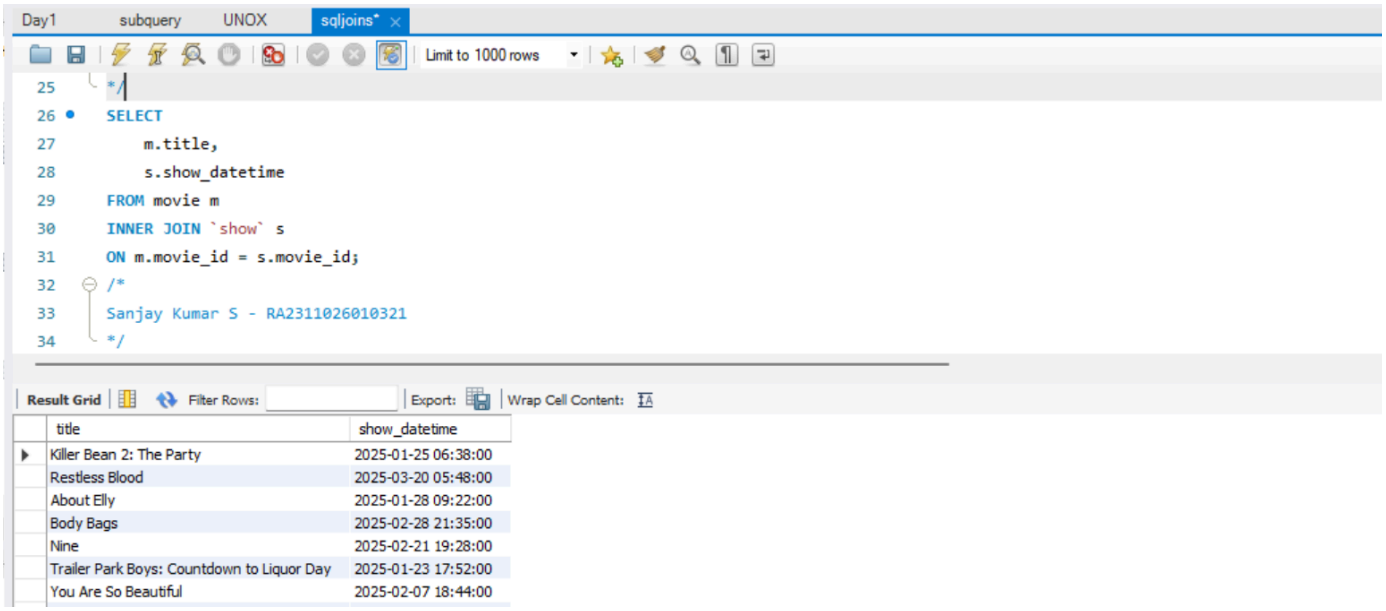
LEVEL 1 – Introduction to Joins & INNER JOIN

Problem 1: List movies and their shows

Complexity: **Easy**

Scenario:

The manager wants a list of all movies along with their scheduled show times.

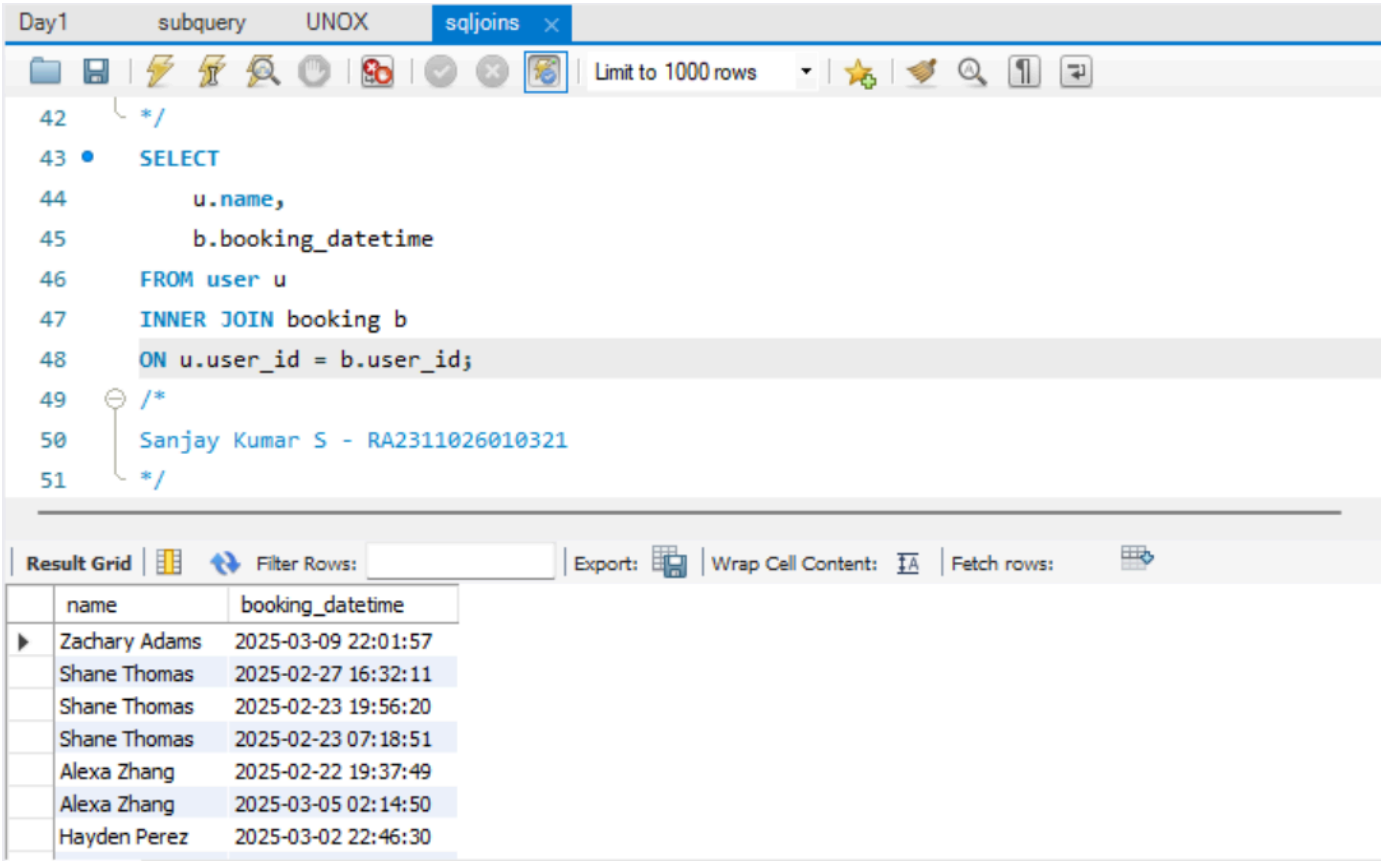


Problem 2: List users and their bookings

Complexity: Easy

Scenario:

Find all users who have made at least one booking, along with booking date.

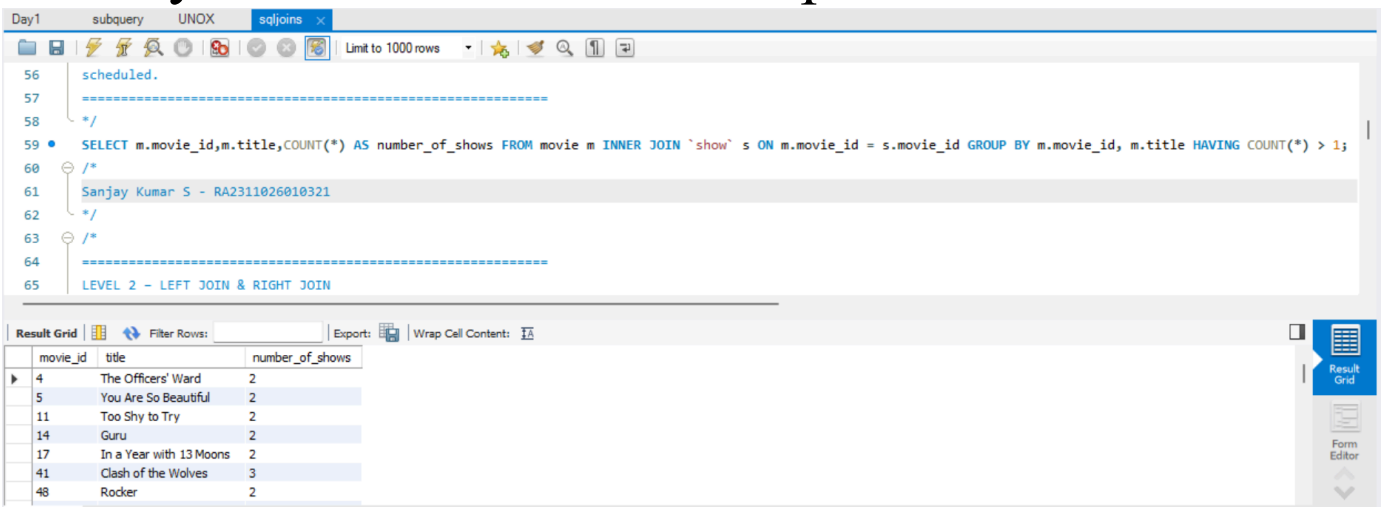


Problem 3 (Higher-order): Movies with more than one show

Complexity: **Medium**

Scenario:

Identify movies that have multiple shows scheduled.



LEVEL 2 – LEFT JOIN & RIGHT JOIN

Problem 1: List all movies and any reviews

Complexity: Easy

Scenario:

Show all movies, even those without reviews.

```sql
/*
============================================================
Problem 1 : List all movies and any reviews
Show all movies, even those without reviews.
============================================================
*/
SELECT m.title, r.review_id, r.comments FROM movie m LEFT JOIN review r ON m.movie_id = r.movie_id;
/*
Sanjay Kumar S - RA2311026010321
*/
```

Result Grid

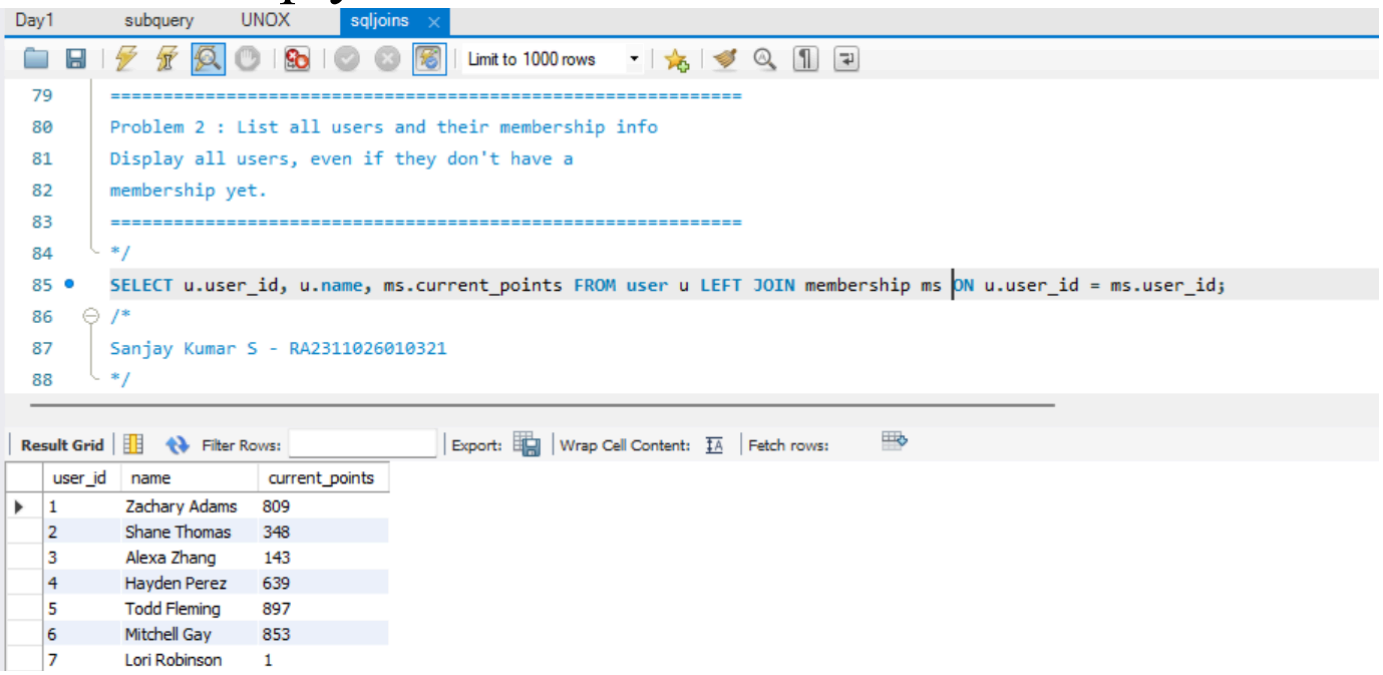
movie_id	title	genre	rating	status	poster_image_url
1	Avatar	Sci-Fi	5.1	Coming Soon	https://placekitten.com/410/120
2	Nine	Comedy	7.5	Now Showing	https://placekitten.com/80/766
3	Hard Luck	Horror	1.4	Now Showing	https://dummyimage.com/718x260
4	The Officers' Ward	Comedy	1.7	Coming Soon	https://dummyimage.com/884x721
5	You Are So Beautiful	Horror	5.9	Coming Soon	https://dummyimage.com/882x449
6	It's Complicated	Horror	9.0	Now Showing	https://picsum.photos/886/659
7	Sherlock Holmes	Drama	2.8	Coming Soon	https://placekitten.com/473/484

Problem 2: List all users and their membership info

Complexity: Easy

Scenario:

Display all users, even if they don’t have a membership yet.

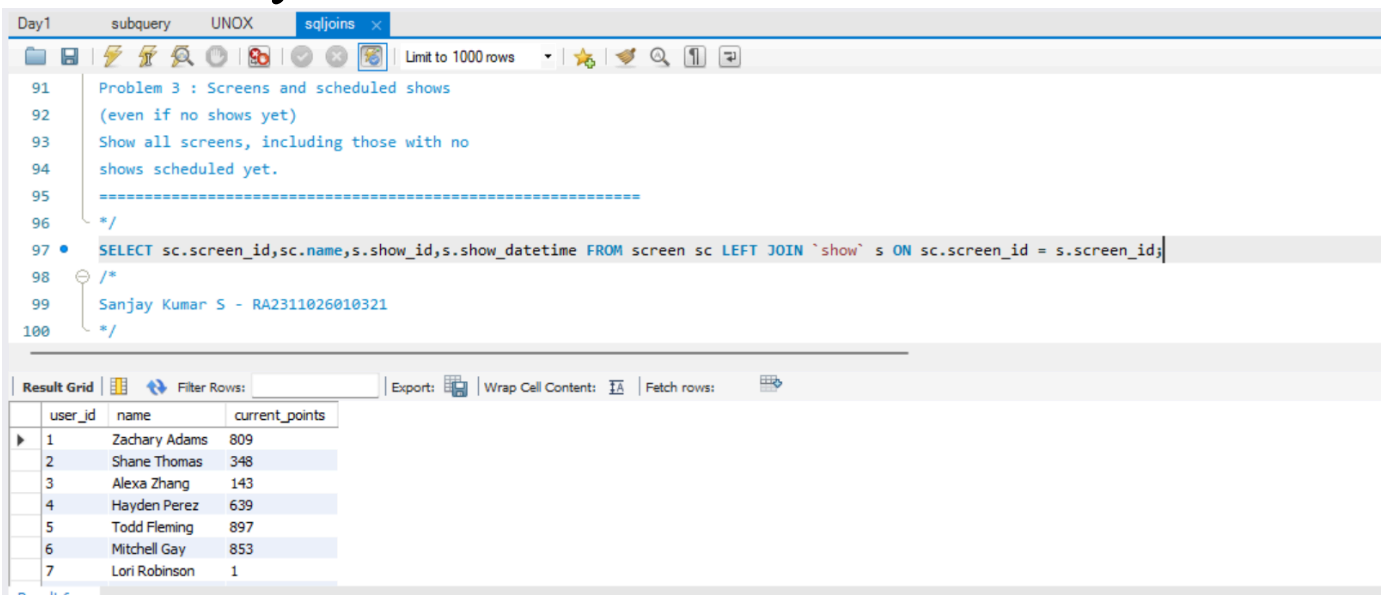


Problem 3 (Higher-order): Screens and scheduled shows (even if no shows yet)

Complexity: Easy

Scenario:

Show all screens, including those with no shows scheduled yet.



Problem 4: Right JOIN Example (Screens and Shows)

Complexity: Easy

Scenario:

Display all shows and their associated screens. Include shows even if screen info is missing.

Day1 subquery UNOX sqljoins

Limit to 1000 rows

```
(Screens and Shows)
Display all shows and their associated
screens. Include shows even if screen
information is missing.
============================================================
*/
SELECT sc.screen_id, sc.name,s.show_id,s.show_datetime FROM screen sc RIGHT JOIN `show` s ON sc.screen_id = s.screen_id;
/*
Sanjay Kumar S - RA2311026010321
*/
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

screen_id	name	show_id	show_datetime
2	B	1	2025-01-25 06:38:00
3	C	2	2025-03-20 05:48:00
4	D	3	2025-01-28 09:22:00
2	B	4	2025-02-28 21:35:00
1	A	5	2025-02-21 19:28:00
3	C	6	2025-01-23 17:52:00
4	D	7	2025-02-07 18:44:00

Result 7

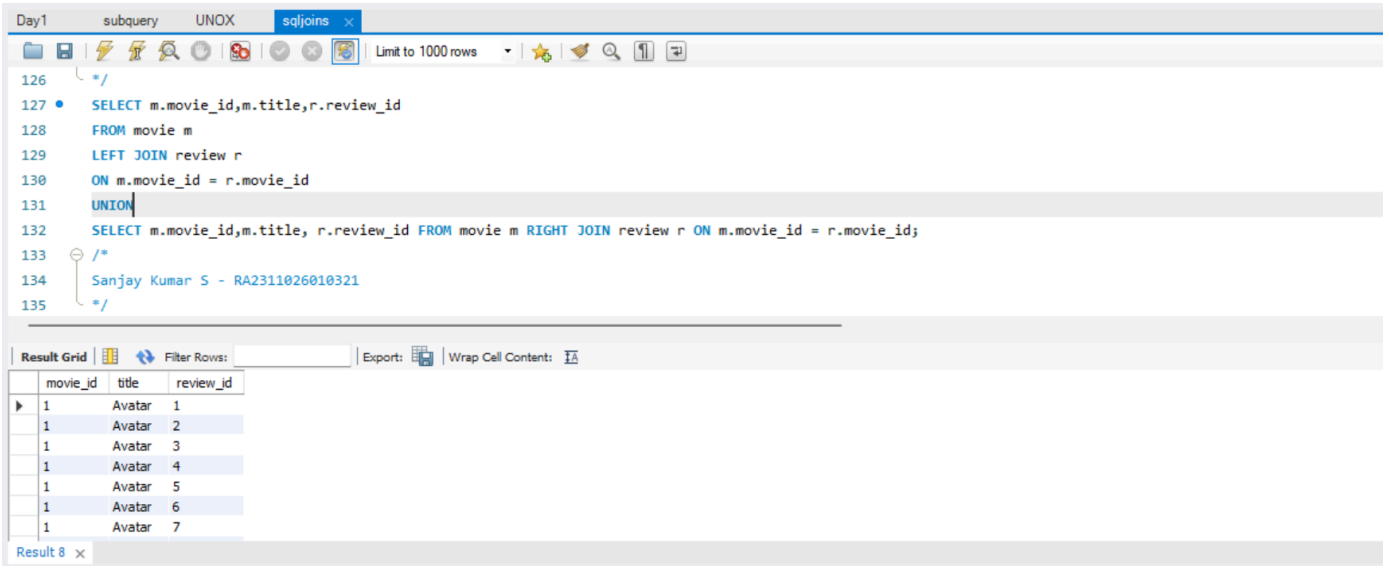
LEVEL 3 – FULL OUTER JOIN & CROSS JOIN

Problem 1: All movies and all reviews

Complexity: Hard

Scenario:

Display all movies and reviews, including movies without reviews and reviews without movies (simulated FULL OUTER JOIN using UNION).

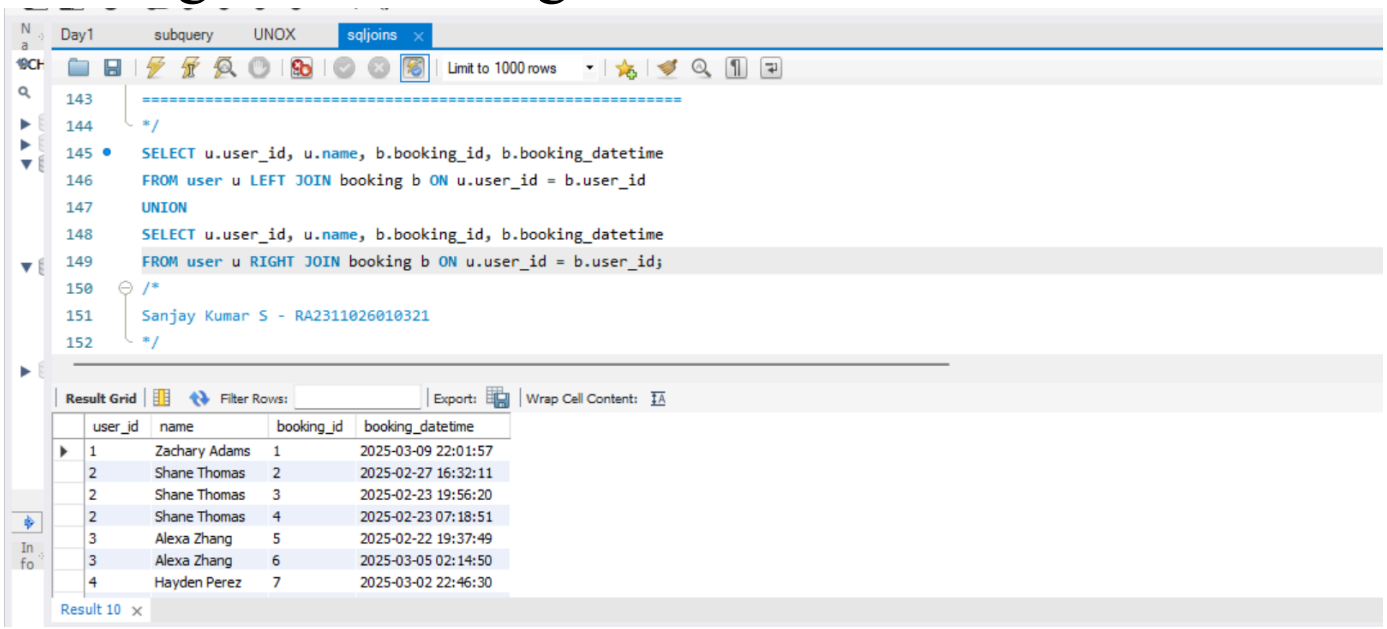


Problem 2: Users and bookings (FULL OUTER JOIN simulation)

Complexity: Hard

Scenario:

List all users and bookings, including users with no bookings and bookings with no user info.



Problem 3 (Higher-order): Generate all seat–show combinations

Complexity: Easy

Scenario:

Generate every possible seat–show combination for planning purposes (CROSS JOIN).